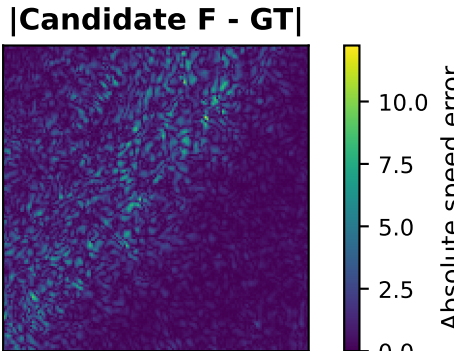
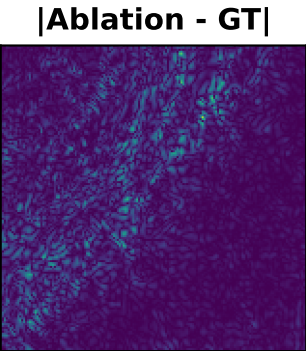
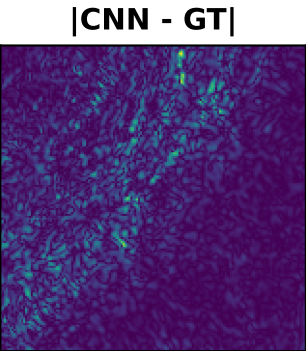
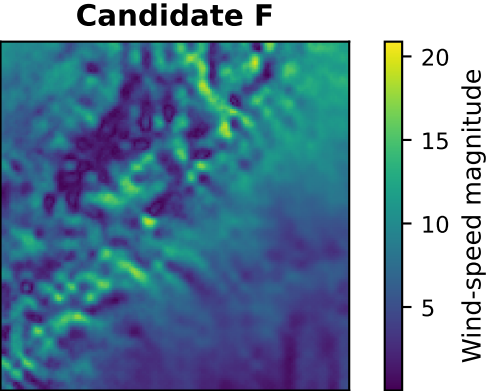
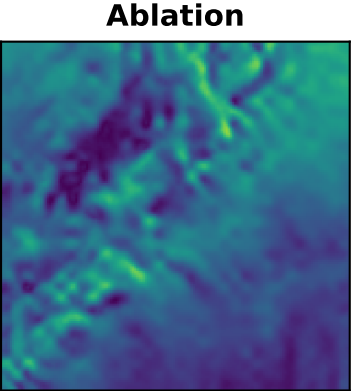
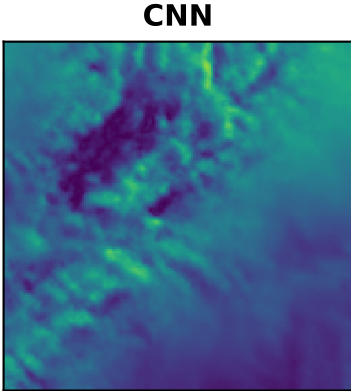
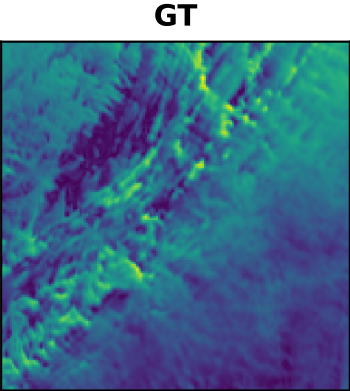
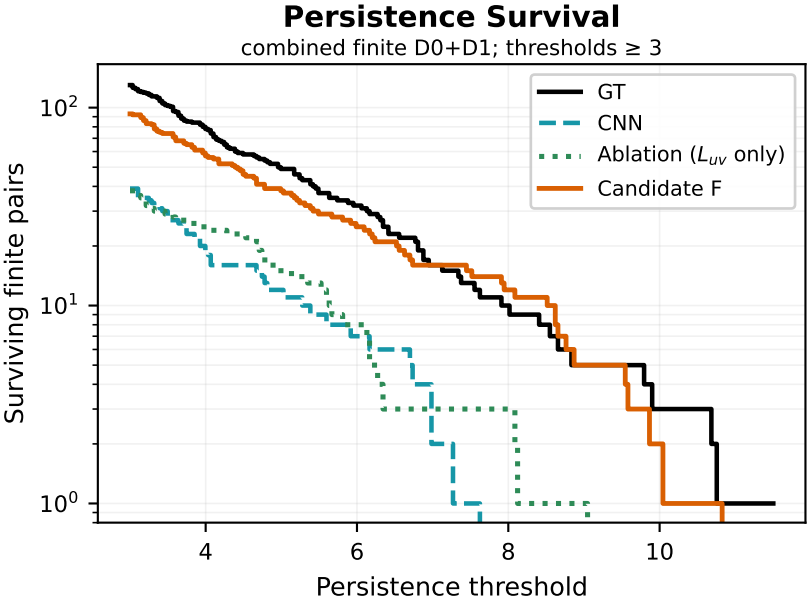
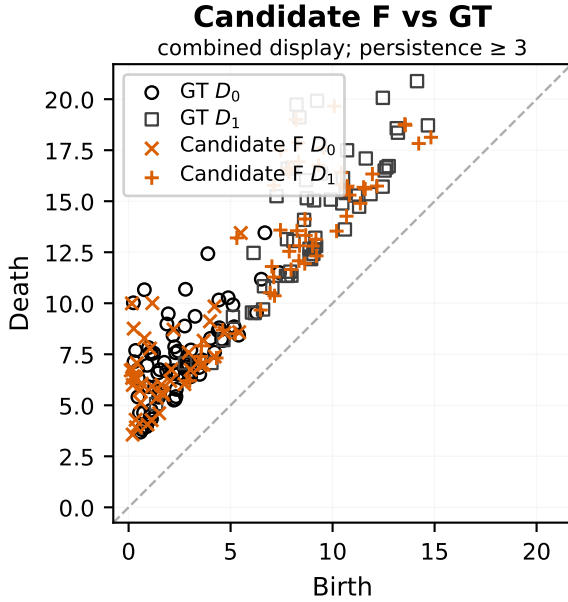
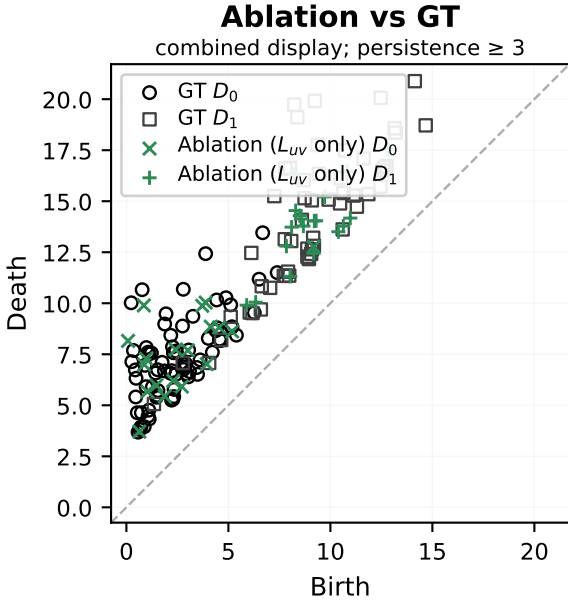
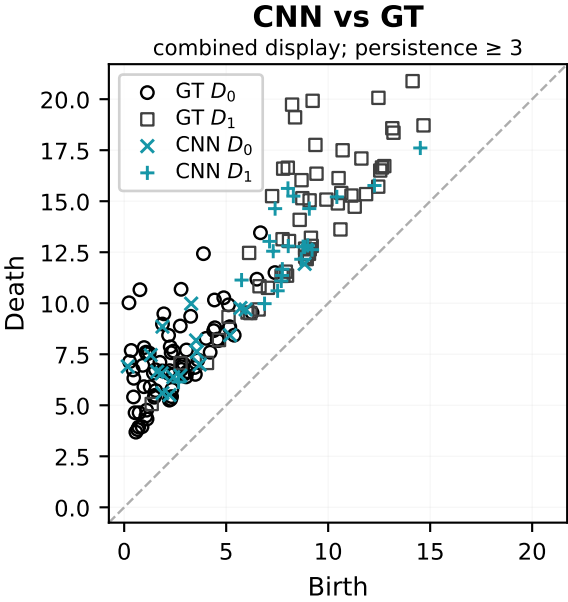


Sample 78: Topology-Aware Gains Beyond Matched Fine-Tuning

CNN = pretrained baseline; Ablation = matched L_{uv} -only fine-tuning; Candidate F = $L_{uv} + L_{grad}$ + repaired E2 supervision.



Topology chain (lower is better)

$W_{2,\infty}$: CNN 23.701 → Abl. 24.152 → F 16.101

$W_{2,2}$: CNN 30.568 → Abl. 31.164 → F 20.145

Conventional fidelity gaps

F vs CNN (original closeness rank 13/168): $|\Delta\text{PSNR}|=0.559$ dB, MAE=9.5%, RMSE=3.5%

F vs L_{uv} ablation: $|\Delta\text{PSNR}|=1.298$ dB, MAE=18.4%, RMSE=16.2%

Matched-control interpretation

Candidate F is lower than the matched L_{uv} -only fine-tuning control under all three validated PD metrics. This sample therefore supports an auxiliary-loss effect beyond ordinary fine-tuning.